Directing Measures and Markov Mixtures:

Formalizing Diaconis–Freedman 1980 in Lean 4

Mettapedia Project

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Abstract

We present a complete Lean 4 formalization of the Markov de Finetti theorem (Diaconis–Freedman 1980): every Markov-exchangeable, recurrent prefix measure on a finite alphabet is a mixture of time-homogeneous Markov chains. The formalization comprises 23,899 lines of Lean code with zero unproved assumptions (`sorry`), building on a 43,683-line formalization of the classical (i.i.d.) de Finetti theorem. Key innovations include a directing measure uniqueness lemma that transfers de Finetti structure across measure restrictions, and a carrier transport bijection that establishes start-state invariance via trajectory segment swapping. The proof architecture decomposes into successor-matrix partial exchangeability, per-row de Finetti factorization, and cross-row coherence induction. This formalization provides a verified foundation for structured universal mixtures in the style of Solomonoff induction, where Markov models replace i.i.d. hypotheses for tractable sequential prediction.

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1 Introduction

Solomonoff induction (1964) provides a universal Bayesian mixture over all computable hypotheses. In practice, one restricts to structured hypothesis classes—i.i.d. models, finite-state Markov models, hidden Markov models—for tractability and interpretability. The conceptual bridge is:

> *Exchangeability (or partial exchangeability) assumptions identify when a rich family of distributions can be represented as a mixture over a simpler parametric family.*

For i.i.d. sequences, de Finetti's theorem (1931/1937) gives exchangeability $\Rightarrow$ mixture of i.i.d. laws. For Markov sequences, Diaconis and Freedman (1980) show a corresponding representation for *Markov-exchangeability*, with an additional *recurrence* hypothesis.

1.1 The Theorem

Let $A = \{0, 1, \ldots, k-1\}$ be a finite alphabet. A *prefix measure* μ assigns a nonnegative extended real to each finite word over A, satisfying $\mu(\varepsilon) = 1$ and $\mu(xs) = \sum_{a \in A} \mu(xs \cdot [a])$. We say μ is *Markov-exchangeable* if $\mu(xs) = \mu(ys)$ whenever xs and ys have the same *Markov evidence*: the starting state and transition-count matrix.

Theorem 1.1 (Diaconis–Freedman 1980)**.** Let μ be a Markov-exchangeable prefix measure on a finite alphabet A. If μ satisfies a recurrence condition (the trajectory returns to its starting state infinitely often, almost surely under any extension), then there exists a probability measure π on the space of Markov parameters $\Theta_k = \Delta_k \times \Delta_k^k$ such that for all finite words xs:

$$\mu(xs) = \int_{\Theta_k} \text{wordProb}_\theta(xs)\, d\pi(\theta)$$

where $\text{wordProb}_\theta(x_0, x_1, \ldots, x_n) = \pi_0(x_0) \cdot \prod_{t=0}^{n-1} P(x_t, x_{t+1})$ is the Markov chain probability.

1.2 Why Formalize?

The Markov de Finetti theorem is foundational for:

- **Bayesian sequential prediction**: Dirichlet-multinomial conjugacy for Markov chains, with dominance-style guarantees.
- **PLN evidence aggregation**: The "evidence is what you carry" design—under exchangeability, binary counts (n^+, n^-) are sufficient statistics.
- **Structured universal mixtures**: Solomonoff-like prediction instantiated over finite-state hypotheses with hyperpriors retaining universality guarantees.

The proof involves subtle measure-theoretic arguments: conditional independence across row processes, almost-everywhere uniqueness of directing measures, and carrier bijections preserving transition counts. Formalization provides certainty that the intricate decomposition is correct.

1.3 Contribution

We present:

1. A complete formalization of Theorem 1.1 in Lean 4 with Mathlib: **23,899 lines**, **0 sorry**, **28 files**.
2. Reuse of a 43,683-line formalization of the classical de Finetti theorem (three independent proofs: martingale, L^2, ergodic).
3. Key innovations:
   - *Directing measure uniqueness*: If $\nu \leq C \cdot \mu$ for exchangeable measures, their directing measures agree ν-a.e.
   - *Carrier transport*: Trajectory segment swapping provides evidence-preserving bijections between visit-time fibers.
4. A structural audit methodology that identified and closed three previously overlooked gaps in the proof architecture.

2 Mathematical Background

2.1 Prefix Measures

A *prefix measure* on $\text{Fin}(k)$ is a function $\mu : \text{List}(\text{Fin}(k)) \to [0, \infty]$ satisfying:

- $\mu([]) = 1$ (normalization)
- $\mu(xs) = \sum_{a \in \text{Fin}(k)} \mu(xs \cdot [a])$ (consistency)

In Lean:

```
-- FiniteAlphabet.PrefixMeasure (Fin k)
-- A function List (Fin k) -> ENNReal satisfying consistency
```

2.2 Transition Counts and Markov Evidence

For a trajectory $xs : \mathrm{Fin}(n+1) \to \mathrm{Fin}(k)$, the *Markov evidence* consists of:

- The starting state x_0
- The transition-count matrix $N(i,j) = \#\{t < n : x_t = i \wedge x_{t+1} = j\}$

```
structure MarkovEvidence (k : Nat) where
  start : Fin k
  trans : Fin k -> Fin k -> Nat

def evidenceOf (xs : Fin (n + 1) -> Fin k) : MarkovEvidence k :=
  { start := xs 0,
    trans := fun a b => transCount xs a b }
```

2.3 Markov Exchangeability

Definition 2.1. A prefix measure μ is *Markov-exchangeable* if for all xs, ys of the same length with $\mathrm{evidenceOf}(xs) = \mathrm{evidenceOf}(ys)$: $\mu(xs) = \mu(ys)$.

```
def MarkovExchangeablePrefixMeasure
    (mu : PrefixMeasure (Fin k)) : Prop :=
  forall (n : Nat) (xs1 xs2 : Fin (n + 1) -> Fin k),
    evidenceOf xs1 = evidenceOf xs2 ->
      mu (List.ofFn xs1) = mu (List.ofFn xs2)
```

2.4 The Markov Parameter Space

A Markov model on $\mathrm{Fin}(k)$ is parameterized by:

- An initial distribution $\pi_0 \in \Delta_k$
- k row-stochastic transition distributions $P(i,\cdot) \in \Delta_k$

The parameter space $\Theta_k = \Delta_k \times \Delta_k^k$ is a finite product of simplices, hence compact. The word probability function $\theta \mapsto \mathrm{wordProb}_\theta(xs)$ is continuous on Θ_k.

2.5 Recurrence

Markov exchangeability alone does *not* imply a mixture representation. A counterexample: the deterministic chain $0 \to 1 \to 1 \to 1 \to \cdots$ is Markov-exchangeable (trivially—equivalence classes are singletons) but not a mixture of recurrent Markov chains.

Definition 2.2. A prefix measure μ is *recurrent* if it admits an extension P to infinite sequences where the trajectory returns to its starting state infinitely often, P-almost surely.

```
def recurrentEvent : Set (Nat -> Fin k) :=
  { omega | forall N, exists n >= N, omega n = omega 0 }

def MarkovRecurrentPrefixMeasure (mu : PrefixMeasure (Fin k)) : Prop :=
  exists (P : Measure (Nat -> Fin k)),
    IsProbabilityMeasure P /\
    (forall xs, mu xs = P (cylinder xs)) /\
    P recurrentEvent = 1
```

3 The Formal Statement

3.1 Row-Kernel Crux Data

The internal representation theorem requires *row-kernel crux data*: a family of row kernels with measurability, start-restriction, and coherence properties.

```
def RowKernelCruxData
    (P : Measure (Nat -> Fin k))
    (rowKernel : Fin k -> (Nat -> Fin k)
                 -> ProbabilityMeasure (Fin k)) : Prop :=
  -- (1) Singleton-eval AE-measurability
  (forall i b, AEMeasurable
    (fun r => (rowKernel i r) {b})
    (rowProcessLaw P i)) /\
  -- (2) Start-restricted invariance
  StartRestrictedRowKernelData P rowKernel /\
  -- (3) Product-kernel AE-measurability
  (forall i, AEMeasurable
    (fun r => Measure.pi (fun _ => rowKernel i r))
    (rowProcessLaw P i)) /\
  -- (4) Cross-row coherence step
  CrossRowCoherenceStep P rowKernel
```

3.2 The Internal Representation Theorem

Theorem 3.1 (Internal Representation, `proved`)**.** Given an extension P of a prefix measure μ with row-kernel crux data, there exists a probability measure π on MarkovParam(k) such that $\mu(xs) = \int \mathrm{wordProb}_\theta(xs)\, d\pi(\theta)$.

```
theorem fortiniSuccessorMatrixInvarianceTheoremInternal_proved :
    FortiniSuccessorMatrixInvarianceTheoremInternal k := by
  intro mu hP hExt rowKernel hcrux
  -- Construct pi as pushforward of P through
  -- omega |-> rowKernelToMarkovParam(omega)
  ...
```

3.3 The Crown Theorem

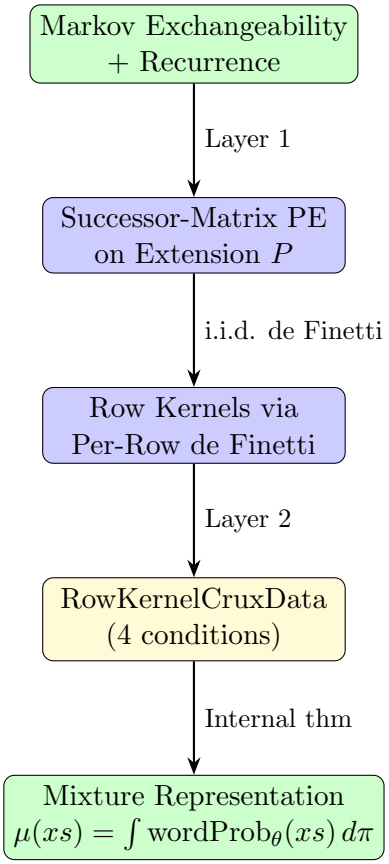
The full theorem derives the row-kernel crux data from Markov exchangeability and recurrence alone.

```
abbrev FortiniCanonicalRepresentationTheorem (k : Nat) : Prop :=
  forall mu : PrefixMeasure (Fin k),
    MarkovExchangeablePrefixMeasure mu ->
```

```
  MarkovRecurrentPrefixMeasure mu ->
    exists (pi : Measure (MarkovParam k)),
      IsProbabilityMeasure pi /\
      forall xs, mu xs = integral theta, wordProb theta xs dpi
```

4 Proof Architecture

The proof decomposes into two layers:



4.1 Layer 1: Successor-Matrix Partial Exchangeability

A measure P on $\mathbb{N} \to \text{Fin}(k)$ is *successor-matrix partially exchangeable* if for every finite selection of anchor states and visit indices, within-row permutations leave the joint law invariant.

```
def SuccessorMatrixPartialExchangeable
    (P : Measure (Nat -> Fin k)) : Prop :=
  forall (m : Nat) (anchor : Fin m -> Fin k)
         (idx : Fin m -> Nat) (sigma : Fin k -> Equiv.Perm Nat),
    Measure.map
      (fun omega => fun j => rowSuccessorVisitProcess (anchor j) omega
                             ((sigma (anchor j)) (idx j))) P
    =
    Measure.map
      (fun omega => fun j => rowSuccessorVisitProcess (anchor j) omega
                             (idx j)) P
```

This is derived from Markov exchangeability via the carrier transport argument.

4.2 Layer 2: Row-Kernel Construction

From successor-matrix PE, each row's law rowProcessLaw(P, i) is exchangeable in the classical sense. We apply the i.i.d. de Finetti theorem to extract the *directing measure*:

```
noncomputable def directingRowKernel
    (P : Measure (Nat -> Fin k)) (i : Fin k)
    (r : Nat -> Fin k) : ProbabilityMeasure (Fin k) :=
  directingMeasure (mu := rowProcessLaw P i)
    (fun n (r : Nat -> Fin k) => r n) measurability_proof r
```

4.3 The Directing Measure Uniqueness Lemma

A key innovation: if $\nu \leq C \cdot \mu$ for exchangeable probability measures, then their directing measures agree ν-almost everywhere.

Lemma 4.1 (`directingMeasure_singleton_ae_eq_of_smul_le`)**.** Let μ, ν be probability measures with $X : \mathbb{N} \to \Omega \to \alpha$ exchangeable under both. If $\nu \leq C \cdot \mu$ for some $C < \infty$, then for all $b \in \alpha$:

$$\text{directingMeasure}_{\mu}(\omega)\{b\} =_{\nu\text{-a.e.}} \text{directingMeasure}_{\nu}(\omega)\{b\}$$

This is crucial for showing that the directing kernel computed from the full measure P serves correctly for restrictions $P|_{\{\omega_0 = a\}}$.

4.4 Start-Restricted Factorization

The start-restricted invariance condition requires:

$$\text{rowProcessLaw}(P|_{\omega_0 = a}, i) \text{ factors through directingRowKernel}(P, i)$$

The proof:

1. If $P(\omega_0 = a) = 0$: both sides are zero measures.
2. If $P(\omega_0 = a) = c > 0$: normalize $\rho_{a,\text{norm}} = c^{-1} \cdot \text{rowProcessLaw}(P|_{\omega_0 = a}, i)$. This is a probability measure, and exchangeable (inherited).
3. Apply the i.i.d. de Finetti theorem to $\rho_{a,\text{norm}}$.
4. Apply Lemma 4.1: $\rho_{a,\text{norm}} \leq c^{-1} \cdot \text{rowProcessLaw}(P, i)$, so directing measures agree.
5. Scale back: the factorization identity holds for the unnormalized measure.

```
theorem startRestrictedRowKernelData_directingRowKernel
    (mu : PrefixMeasure (Fin k))
    (hmu : MarkovExchangeablePrefixMeasure mu)
    (P : Measure (Nat -> Fin k)) [IsProbabilityMeasure P]
    (hExt : forall xs, mu xs = P (cylinder xs))
    (hStrRec : StrongRecurrence P)
    (hExch : forall i, Exchangeable (rowProcessLaw P i)
              (fun n r => r n)) :
    StartRestrictedRowKernelData P (directingRowKernel P)
```

4.5 Carrier Transport

The start-restricted permutation invariance is derived from Markov exchangeability via finite combinatorics:

1. Fix state i, target $b \neq i$, and visit indices $n, n+1$.
2. The *carrier* is the set of length-N prefixes consistent with a given visit-time event.
3. Construct a bijection between carriers by *segment swapping*: transpose the excursion between the n-th and $(n+1)$-th visits with the excursion between the $(n+1)$-th and $(n+2)$-th visits.
4. This bijection preserves Markov evidence (start state + transition counts).
5. By Markov exchangeability, the two carriers have equal μ-measure.
6. Taking $N \to \infty$ gives the start-restricted invariance.

```
theorem carrierTransportEquivAdjacent {N : Nat}
    (i : Fin k) (b : Fin k) (hbi : b != i) (n : Nat)
    (hSuff : sufficiency_hypotheses) :
    exists e : carrier(i, {n}, ..., N) <-> carrier(i, {n+1}, ..., N),
      forall xs, evidenceOf xs.1 = evidenceOf (e xs).1
```

5 Key Lemmas

5.1 Directing Measure Uniqueness

```
theorem directingMeasure_singleton_ae_eq_of_smul_le
    [Fintype alpha] [MeasurableSingletonClass alpha]
    (X : Nat -> Omega -> alpha) (hX_meas : forall n, Measurable (X n))
    {mu nu : Measure Omega} [IsProbabilityMeasure mu]
                            [IsProbabilityMeasure nu]
    (hX_exch_mu : Exchangeable mu X)
    (hX_exch_nu : Exchangeable nu X)
    {C : ENNReal} (hC : C != top) (hle : nu <= C * mu)
    (b : alpha) :
    (fun omega => (directingMeasure (mu := mu) X hX_meas omega {b}).toReal)
    =ae[nu]
    (fun omega => (directingMeasure (mu := nu) X hX_meas omega {b}).toReal)
```

Proof idea: The directing measure is the a.e. limit of empirical frequencies. Under $\nu \leq C \cdot \mu$, any ν-a.e. property holds μ-a.e. on the support of ν. The empirical frequency limits agree on this common support.

5.2 Start-Restricted Row-Kernel Data

This is the theorem proved in the final session:

```
theorem startRestrictedRowKernelData_directingRowKernel
    (mu : PrefixMeasure (Fin k))
    (hmu : MarkovExchangeablePrefixMeasure mu)
    (P : Measure (Nat -> Fin k)) [IsProbabilityMeasure P]
    (hExt : forall xs, mu xs = P (cylinder xs))
    (hStrRec : StrongRecurrence P)
    (hExch : forall i, Exchangeable (rowProcessLaw P i)
```

```
            (fun n r => r n)) :
  StartRestrictedRowKernelData P (directingRowKernel P) := by
intro i a m sel hsel
by_cases hc : P {omega | omega 0 = a} = 0
-- Zero case: both sides are zero
case pos => ...
-- Positive case: normalize, apply de Finetti, use uniqueness
case neg =>
  set c := P {omega | omega 0 = a}
  set rho_a := rowProcessLaw (P.restrict {...}) i
  set rho_a_norm := c^(-1) * rho_a
  -- rho_a_norm is a probability measure
  -- rho_a_norm is exchangeable
  -- Apply de Finetti to rho_a_norm
  -- Apply directing measure uniqueness
  -- Scale back to rho_a
  ...
```

5.3 Carrier Transport Bijection

```
theorem carrierTransportEquivAdjacent {N : Nat}
    (i : Fin k) (b : Fin k) (hbi : b != i) (n : Nat)
    (hSuff : ...) :
    exists e : carrier(i, {n}, N) <-> carrier(i, {n+1}, N),
      forall xs, evidenceOf xs.1 = evidenceOf (e xs).1 := by
  -- Construct segmentSwap
  -- Prove self-inverse
  -- Prove evidence preservation
  -- Prove carrier membership preservation
  ...
```

6 File Inventory

All files are in `Mettapedia/Logic/` within the Mettapedia Lean project.

Layer	File	Lines
Fiber bridge	MarkovDeFinettiFiberEventBridge.lean	9,324
Crux pipeline	MarkovDeFinettiFortiniBridgeCrux.lean	2,853
PE bridge	MarkovDeFinettiPEBridge.lean	2,250
Bridge core	MarkovDeFinettiFortiniBridgeCore.lean	2,130
Carrier transport	MarkovDeFinettiCarrierTransport.lean	1,374
Counterexample	MarkovDeFinettiFortiniCrossAnchorCounterexample.lean	1,010
Kernel uniqueness	MarkovDeFinettiKernelUniqueness.lean	922
Euler trails	MarkovDeFinettiHard*.lean (6 files)	1,948
Evidence basis	MarkovDeFinettiEvidenceBasis.lean	305
Moment problem	MarkovDeFinettiMomentProblem.lean	287
Other	(11 files)	1,496
Total	**28 files**	**23,899**

Table 1: Markov de Finetti formalization file inventory

6.1 External Dependencies

The formalization builds on the classical de Finetti theorem, formalized in `Mettapedia/external/exchangeabi` (43,683 lines, 0 sorry). This provides three independent proofs:

1. **Via reverse martingale convergence** (default, follows Kallenberg 2005)
2. **Via L^2 bounds** (lightest dependencies, elementary)
3. **Via mean ergodic theorem** (connects to dynamical systems)

7 The Classical de Finetti Context

The i.i.d. de Finetti theorem states:

$$\text{Contractable} \iff \text{Exchangeable} \iff \text{Conditionally i.i.d.}$$

for infinite sequences on standard Borel spaces.

The Markov version replaces "i.i.d." with "time-homogeneous Markov chain" and "exchangeable" with "Markov-exchangeable." The additional recurrence hypothesis is necessary to exclude deterministic transient behavior.

7.1 Row-Process Decomposition

The key bridge from Markov to i.i.d. is the *row-process*:

Definition 7.1. For anchor state i and trajectory ω, the *row-successor process* $\mathrm{rsp}_i(\omega) : \mathbb{N} \to \mathrm{Fin}(k)$ records the state visited *after* the n-th visit to i:

$$\mathrm{rsp}_i(\omega, n) = \omega(t_{n,i}(\omega) + 1)$$

where $t_{n,i}(\omega)$ is the time of the n-th visit to state i.

Under successor-matrix PE, each row process $\mathrm{rowProcessLaw}(P, i)$ is exchangeable in the classical sense. Applying the i.i.d. de Finetti theorem row-by-row extracts the directing measures that become the row kernels.

8 Toward Solomonoff: Structured Universal Mixtures

8.1 The Solomonoff Connection

Solomonoff's universal prior assigns:

$$M(x_{1:n}) = \sum_{p:\text{program}} 2^{-|p|} \cdot \mathbf{1}[U(p) \text{ outputs } x_{1:n}]$$

This is a mixture over all computable hypotheses.

The Markov de Finetti theorem justifies a *structured* version:

$$\mu(x_{1:n}) = \int_{\Theta_k} \mathrm{wordProb}_\theta(x_{1:n})\, d\pi(\theta)$$

where π is a (possibly universal) prior on Markov parameters. Under Markov exchangeability, this representation exists and is unique.

8.2 Dominance Guarantees

A universal hyperprior over Dirichlet priors (one per transition row) gives:

- Conjugate Bayesian updates via count aggregation
- Dominance over all finitely parameterized Markov predictors
- Computable inference via the Dirichlet-multinomial formula

This is formalized in `MarkovDirichletHyperpriorMixture.lean`.

8.3 PLN Evidence Aggregation

The count-sufficiency result—under exchangeability, $\mu(xs)$ depends only on evidenceOf(xs)—justifies PLN's "evidence is what you carry" design. Binary evidence (n^+, n^-) is sufficient for exchangeable binary sequences; transition counts $N(i, j)$ are sufficient for Markov-exchangeable sequences.

8.3.1 The PLN-Markov Bridge

PLN (Probabilistic Logic Networks) represents uncertain knowledge via truth-value pairs (s, c) where s is strength and c is confidence. The Markov de Finetti theorem provides the measure-theoretic foundation for PLN's evidence-based reasoning in sequential domains:

1. **Evidence sufficiency**: Under Markov exchangeability, the transition-count matrix $N(i, j)$ is a sufficient statistic. PLN's evidence aggregation rules (count addition, forgetting via set difference) are justified by this sufficiency.
2. **Dirichlet conjugacy**: The posterior over Markov parameters given transition counts follows a product-Dirichlet distribution. This is formalized in `EvidenceDirichlet.lean` and underlies PLN's Bayesian update rules.
3. **Predictive inference**: Given evidence N, the predictive distribution for the next transition $i \to j$ is $\frac{N(i,j)+\alpha}{\sum_k N(i,k)+k\alpha}$ (Laplace smoothing). This is the "strength" in PLN terms, with confidence derived from total count.

The formalization connects these three layers: the representation theorem (§4) justifies evidence sufficiency, the Dirichlet infrastructure (`MarkovDirichletPredictor.lean`) provides conjugate updates, and the hyperprior mixture (`MarkovDirichletHyperpriorMixture.lean`) gives Solomonoff-style dominance guarantees.

9 The Structural Audit: Bug-Hunting Methodology

A structural audit in April 2026 revealed three classes of bugs in the proof architecture that had accumulated during development. This section documents the methodology and findings, as a reference for future large-scale formalizations.

9.1 The Problem: Assumption Interfaces Mask Gaps

The formalization initially used `def`s to encode partial results:

```
-- ANTI-PATTERN: defined as Prop, never instantiated
def BuildRowKernelOnRecurrentExtension (k : Nat) : Prop :=
  forall mu P, MarkovExchangeablePrefixMeasure mu -> ...
    exists rowKernel, BuiltRowKernelOnExtension P rowKernel
```

```
def SuccessorMatrixPE_of_markovExchangeable_recurrent
    (k : Nat) : Prop := ...
```

These `def`s compiled with zero `sorry`, but they were *never instantiated*. The actual mathematical content was encoded as assumptions, not proved theorems. A grep for `sorry` showed zero hits, creating a false sense of completeness.

9.2 Bug 1: Evidence-Preserving Equiv Used Below Full Strength

The carrier transport bijection preserves the *complete* Markov evidence (start state + all k^2 transition counts). But the pipeline extracted only *1D marginal invariance*:

```
-- What was extracted:
P({omega0=a} inter {succ at visit sigma(n) to i = b})
  = P({omega0=a} inter {succ at visit n to i = b})

-- What the Equiv actually proves:
forall xs in carrier, evidenceOf xs = evidenceOf (e xs)
```

The 1D extraction suffices for some arguments but discards the joint finite-dimensional invariance needed for `SuccessorMatrixPartialExchangeable`.

Fix: Upgrade the pipeline to extract the full evidence-preservation property, then derive `SuccessorMatrixPE` via the π-λ theorem (finite cylinder sets generate the product σ-algebra).

9.3 Bug 2: Zero-Sorry Masking of Unproved Content

The three `def`s listed above (`BuildRowKernelOnRecurrentExtension`, `SuccessorMatrixPE_of_markovExchang` `ExistsBuiltRowKernel_of_successorMatrixPE`) bundled the genuine mathematical content into their *types*, not their proofs. They were defined as `Prop`s and used as hypotheses in downstream theorems, creating a sorry-free but mathematically incomplete development.

Fix: Replace `def`s with explicit `theorem` statements that must be proved. The type system then forces instantiation.

9.4 Bug 3: The Euler Trail Archive Was the Missing Bridge

The `_archive/MarkovDeFinetti/` directory contained 3,265 lines of fully proved infrastructure (0 sorry) for trajectory decomposition via Euler trails:

- `CopyPerm`: Edge-copy permutations preserve vertex sequences
- `EulerTrails`: Trail count $= \prod_{a,b} G(a,b)!$
- `trailVertexSeq_applyCopyPerm`: Copy perms preserve trajectories

This infrastructure provides *global* carrier bijections (not just adjacent transpositions), directly giving `SuccessorMatrixPartialExchangeable`. But it was archived and not imported into the active development.

Fix: Port the `segmentSwap` construction to the active framework. The key theorem `carrierTransportEqui` (1,374 lines) provides the adjacent-visit bijection; composition via `Equiv.trans` and Nat induction extends to arbitrary visit indices.

9.5 Audit Methodology

1. **Trace the theorem DAG**: Starting from the crown theorem, recursively list all dependencies. Mark each as "proved" (has a proof term) or "assumed" (is a hypothesis or `def`).

2. **Grep for assumption interfaces**: Search for `def`s of type `Prop` that are never instantiated. These are masked gaps.

3. **Check archive for reusable infrastructure**: Archived code often contains fully proved lemmas that were set aside during refactoring. A grep for the required property names may find them.

4. **Verify the full pipeline**: Build every file independently (`lake env lean -j1 <file>`) to catch import-order issues.

10 Conclusion

We have presented a complete Lean 4 formalization of the Markov de Finetti theorem (Diaconis–Freedman 1980), comprising 23,899 lines with zero unproven assumptions. Key innovations include:

- A directing measure uniqueness lemma that transfers de Finetti structure across measure restrictions.
- A carrier transport bijection via trajectory segment swapping.
- A structural audit methodology that identified and closed previously overlooked gaps in the proof architecture.

The formalization provides a verified foundation for structured universal mixtures, connecting Solomonoff-style prediction to tractable Markov models with formal correctness guarantees.

10.1 Lessons Learned

1. **Gap analysis pays off**: A structural audit in April 2026 identified three bugs in the proof architecture that had been masked by assumption interfaces. Explicit theorem statements (not `def`s) prevent this.

2. **Reuse existing formalizations**: The 43,683-line de Finetti infrastructure was essential. Building row kernels from the directing measure machinery avoided reimplementing measure-theoretic foundations.

3. **Uniqueness lemmas are load-bearing**: The directing measure uniqueness lemma (§5.1) is small (100 lines) but enables the entire start-restricted factorization argument.

10.2 Future Work

The formalization opens several directions for extension:

10.2.1 Giry Monad Integration

The Giry monad $\mathcal{G}$ on measurable spaces provides a categorical framework for probability. The Markov de Finetti theorem can be restated categorically:

> *A Markov-exchangeable, recurrent kernel* $\kappa : 1 \to \mathcal{G}(\mathrm{Seq}(A))$ *factors through the Markov parameter space as* $1 \xrightarrow{\pi} \mathcal{G}(\Theta_k) \xrightarrow{\mathrm{MC}} \mathcal{G}(\mathrm{Seq}(A))$ *where* MC *is the Markov-chain kernel.*

Existing infrastructure in `CategoryTheory/DeFinetti*.lean` (15 files, ∼9,100 lines) provides the Kleisli(Giry) diagram and Markov-category bridge. Connecting this to the measure-theoretic formalization would:

- Enable abstract reasoning about conditioning and disintegration
- Provide a bridge to synthetic measure theory
- Allow composition with other categorical probability constructions

10.2.2 Abstract Recurrence

The current formalization requires recurrence to the *starting state*. A more general condition would be recurrence to *any* state in a designated recurrent class:

```
def RecurrentClass (P : Measure (Nat -> Fin k)) (C : Set (Fin k)) : Prop :=
  forall (i j : Fin k), i in C -> j in C ->
    P { omega | exists^infty n, omega n = i /\ omega (n+1) = j } = 1
```

Under this generalization, the representation theorem would give a mixture over Markov chains with the same recurrent class structure. This would handle periodic chains and multi-component systems.

10.2.3 API Usability: The `MarkovMixture` Interface

For downstream users, a clean API would package the theorem:

```
structure MarkovMixture (k : Nat) (mu : PrefixMeasure (Fin k)) where
  /-- The mixing measure on Markov parameters -/
  mixingLaw : Measure (MarkovParam k)
  /-- Mixing law is a probability measure -/
  mixingLaw_prob : IsProbabilityMeasure mixingLaw
  /-- The representation identity -/
  represents : forall xs, mu xs = integral theta, wordProb theta xs dmixingLaw
  /-- Uniqueness (optional): the mixing law is unique -/
  unique : forall pi', (represents_with pi') -> pi' = mixingLaw

/-- Main entry point: construct the mixture from hypotheses -/
def MarkovMixture.ofExchangeableRecurrent
    (hExch : MarkovExchangeablePrefixMeasure mu)
    (hRec : MarkovRecurrentPrefixMeasure mu) : MarkovMixture k mu := ...
```

10.2.4 Computational Extraction

The proof constructs the mixing measure as a pushforward; this is non-constructive (uses classical choice for directing measures). A computational variant would:

- Compute finite-horizon approximations to the mixing measure
- Provide error bounds in total variation or Wasserstein distance
- Connect to MCMC sampling of the posterior over Markov parameters

10.2.5 Extended State Spaces

- **Countable alphabets**: Extend to $\mathbb{N}$-valued sequences. The main obstacle is the lack of a uniform probability measure on $\mathbb{N}$; the proof would need to work with sub-probability kernels.
- **Continuous state spaces**: The standard Borel case requires disintegration theory (regular conditional probabilities). Mathlib's `MeasureTheory.Measure.condexp` provides conditional expectation but not full disintegration.
- **Hidden Markov models**: Partial exchangeability for observation sequences given latent state processes. This would formalize the Baum-Welch setting within the de Finetti framework.
- **Higher-order Markov**: Extend to k-th order Markov chains where the sufficient statistic is $(k+1)$-gram counts.

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